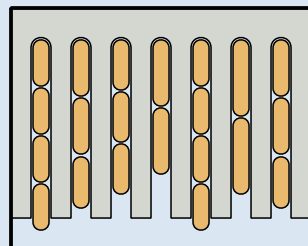
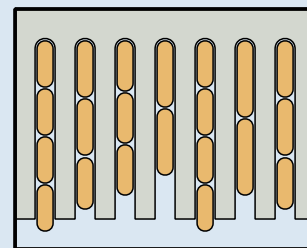


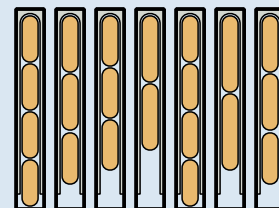
rotation
and drift
correction



U-Net
ROI model

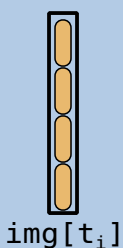


crop and
resize

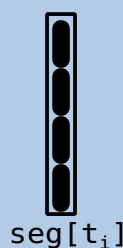


For each ROI (k)

For each frame (time t_i)



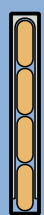
U-Net
segmentation
model



For each cell in frame t_{i-1} (cell j)



$img[t_i]$



$seg[t_{i-1}][j]$



$seg[t_i]$

U-Net
tracking
model



tracking scores
for cell j

attributions
between
 t_{i-1} and t_i



$labels[t_i]$

- computation of cell features (length, area, fluorescence)
- building lineage tree
- computation of growth rate